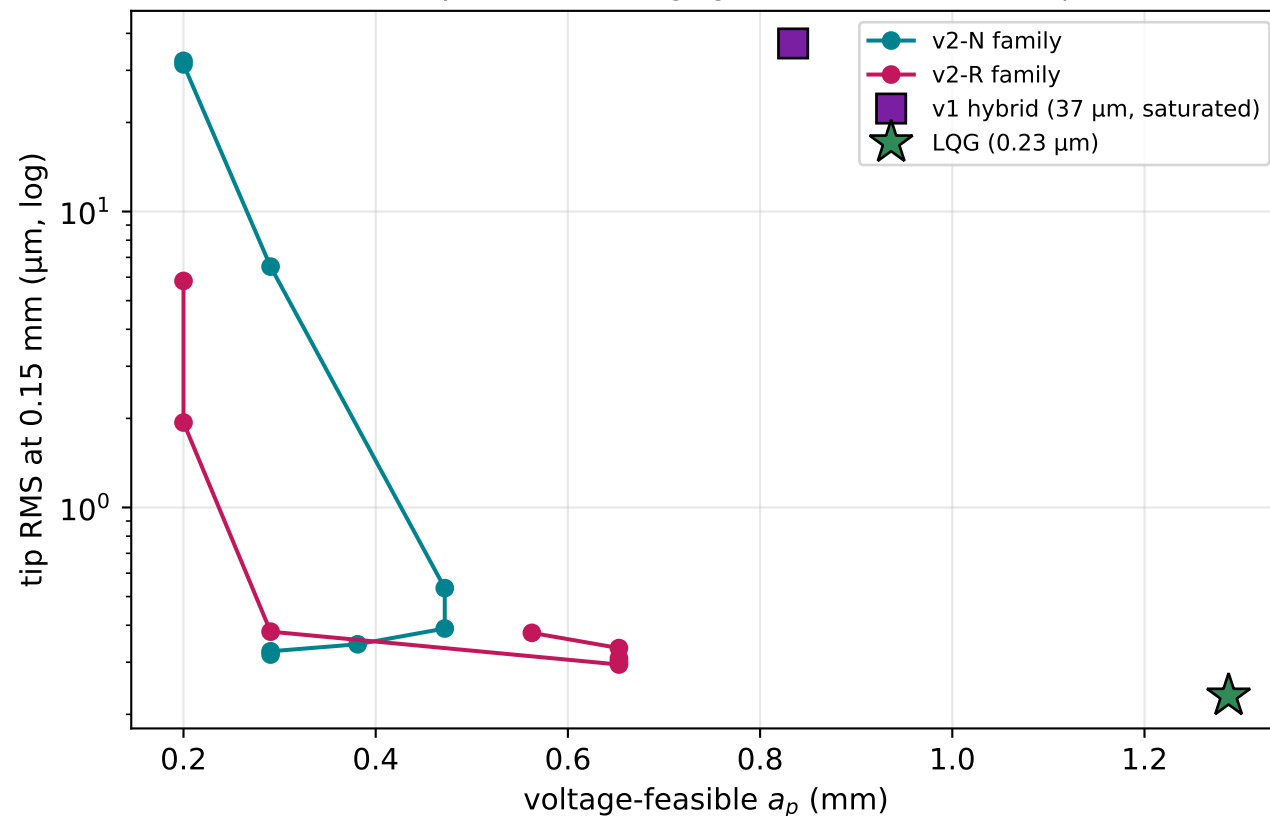
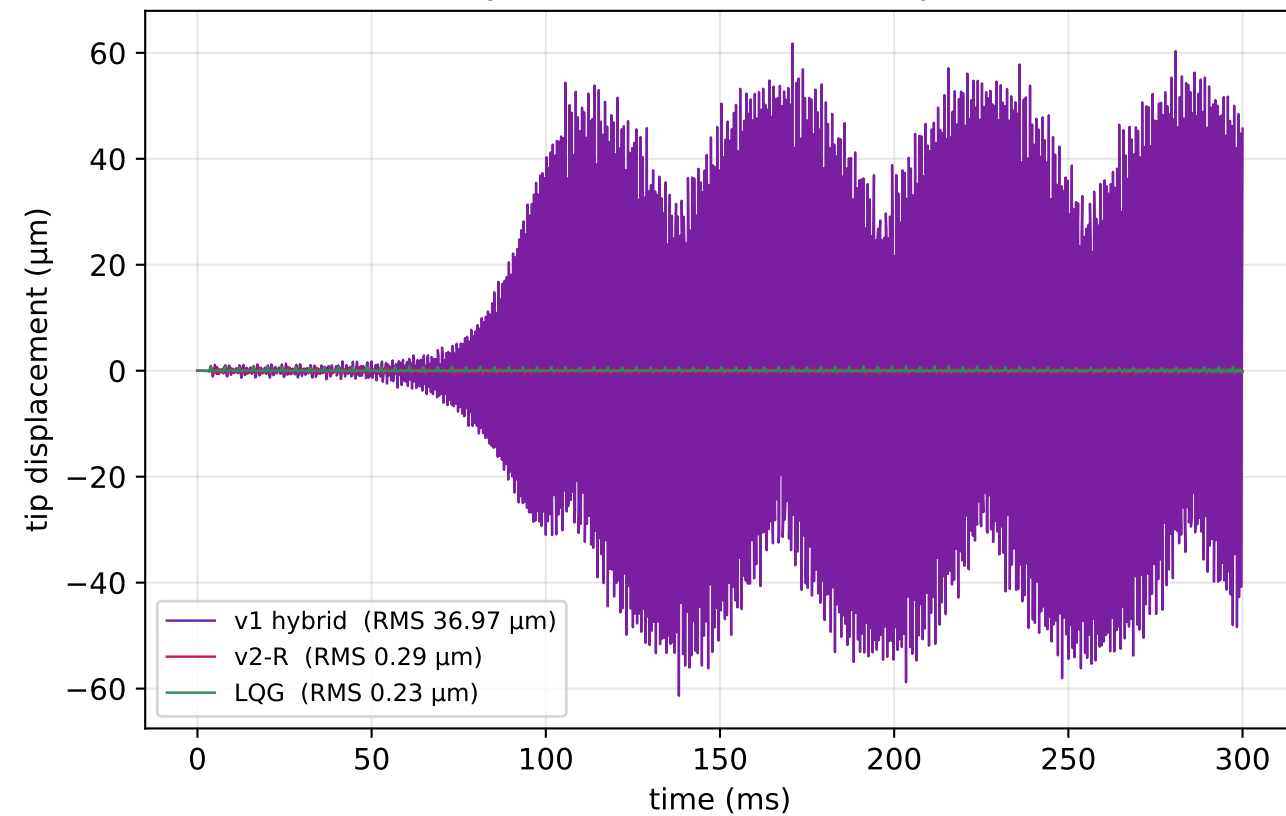


HYBRID v2 improvements: ESO leakage + injection roll-off + robust min-max co-design (tip sensor)

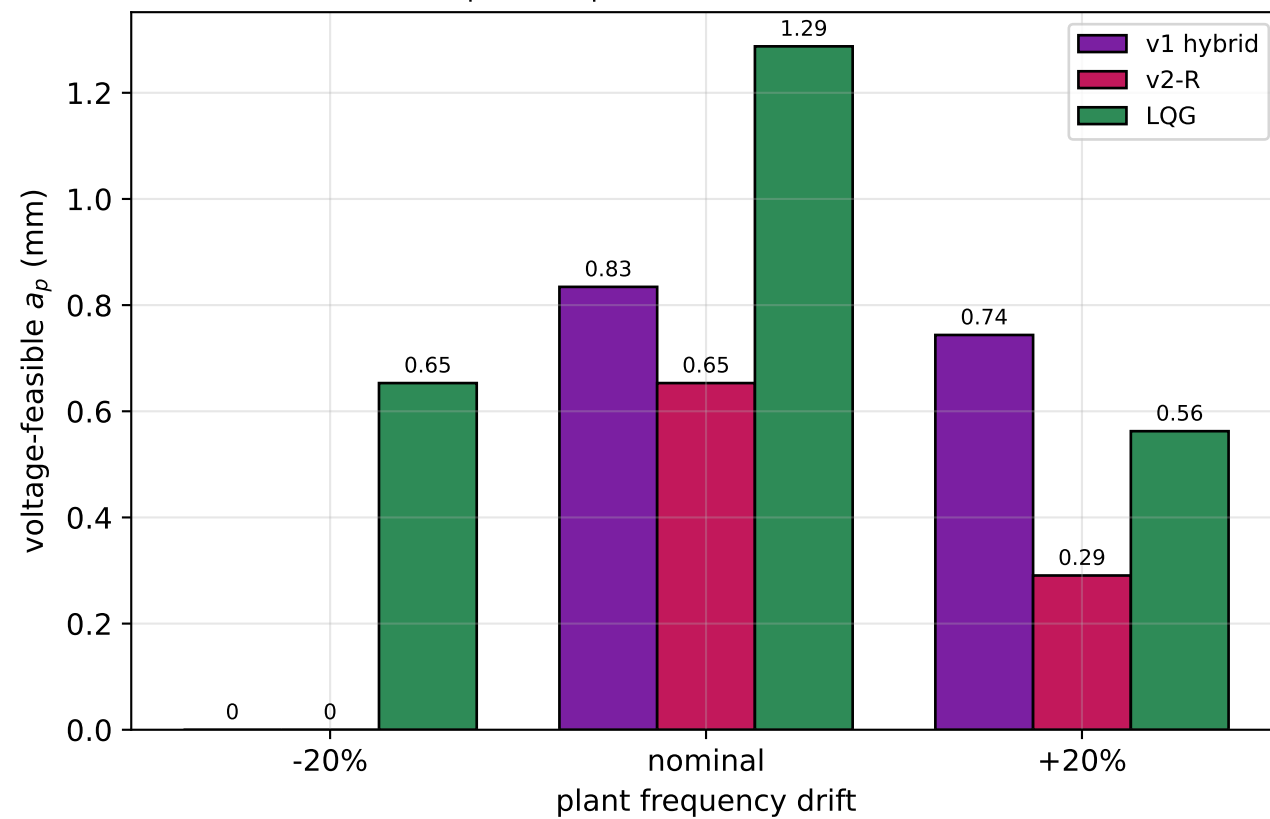
(a) margin vs finish (gain-scale sweeps, non-monotone: the limit cycle returns at high gains); v1 = saturated 37 μm



(b) rejection at $a_p = 0.15$ mm on the tip sensor
ESO leakage removes the saturated limit cycle ($\sim 125\times$)



(c) drift robustness: the -20% weakness persists on the refined plant despite the linear min-max certificate



(d) spillover repair (refined-plant linear boundary 0.01 $\rightarrow$ 0.96 mm) and load-bearing ablation

